

Noise Component Analysis of ePMR and HAMR across radius for Areal Density Improvements

Amirhossein Sayyafan¹, Jonas Goode¹, Roger Wood², *Life Fellow, IEEE*, Rick Galbraith³,
Niranjay Ravindran³, Pradhan Bellam³, and Austin Striegel³

¹Western Digital Corporation, Irvine, California, USA, amir.sayyafan@wdc.com, jonas.goode@wdc.com

²Western Digital Corporation, San Jose, California, USA, roger.wood@wdc.com

³Western Digital Corporation, Rochester, Minnesota, USA, rick.galbraith@wdc.com, niranjay.ravindran@wdc.com,
pradhan.bellam@wdc.com, austin.striegel@wdc.com

As magnetic recording is pushed to ever higher areal densities, a detailed breakdown of the various noise sources becomes essential in understanding and optimizing future systems. Two hard disk drives are examined: one employing energy-assisted Perpendicular Magnetic Recording (ePMR) and the other, Heat-Assisted Magnetic Recording (HAMR). Measurements are made at three radii (ID, MD, OD), characterized by large changes in head-skew and media velocity. The ‘noise’ includes inter-track interference (ITI) from the two adjacent tracks, noise from the recording medium (divided into transition-noise and non-transition-noise), and noise that arises from the head and electronics. The latter is relatively flat (white) and provides a good reference point allowing comparisons across the drives and test radii. Such a study is valuable in exploring potential gains from advanced signal processing.

Index Terms— Hard Disk Drives, Heat-Assisted Magnetic Recording, energy-Assisted Perpendicular Magnetic Recording, Shingled Magnetic Recording, Conventional Magnetic Recording

I. INTRODUCTION

TO CONTINUE growth of hard disk drive (HDD) areal density requires both advances in recording technology and a clear understanding of how the read channel responds under different operating conditions. In heat-assisted magnetic recording (HAMR) systems, measured and modeled responses at ultra-high areal densities show that the relationship among media response, waveform shape, and channel behavior becomes increasingly important as recording density increases [1]. At the system level, advanced channel coding and shingled magnetic recording (SMR) operation provide additional paths for extending HDD scaling, but they also place greater demands on the signal processing chain used to recover data from closely spaced tracks [2]. These trends motivate the use of noise component analysis to look beyond total readback noise and understand which parts of the system are limiting performance. In this paper, we examine HAMR and energy-assisted perpendicular magnetic recording (ePMR) across inner diameter (ID), middle diameter (MD), and outer diameter (OD), with SMR as the focus and conventional magnetic recording (CMR) data reserved for later comparison. By comparing the separated noise contributions across radius and technology, it can help identify where signal processing improvements, read-channel tuning, or radius-aware optimization may provide additional margin and support further areal density improvement.

II. SIGNAL PROCESSING CHAIN FOR NOISE COMPONENT ANALYSIS

Fig. 1 shows the source flow used to prepare measured readback waveforms for noise component analysis. A brick-wall continuous time filter (CTF) is first applied to the raw waveforms captured from the drive to impose a controlled bandwidth and place all measurements under a consistent

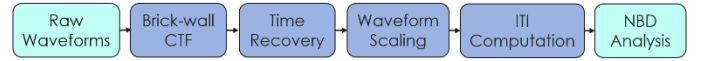


Fig. 1. Block diagram of source flow

effective channel response. The waveform then passes through time recovery, which aligns the samples to the correct symbol timing so that timing offsets are not misinterpreted as noise. Waveform scaling accounts for normalizing amplitude differences caused by variations in gain, radius, data rate, or recording technology. Finally, the inter-track-interference (ITI) computation estimates the portion of the conditioned waveform caused by adjacent track coupling, allowing the final noise component analysis block to treat ITI as a separate contribution rather than mixing it with media or electronic noise. This processing sequence is consistent with the need for carefully conditioned readback responses when evaluating high-density magnetic recording channels.

Fig. 2 explains how the conditioned waveform is decomposed into electronic, ITI, transition, and non-transition noise powers. The analysis is performed on the central track N. Re-Write #1 and Re-Write #2 use the same nominal data pattern

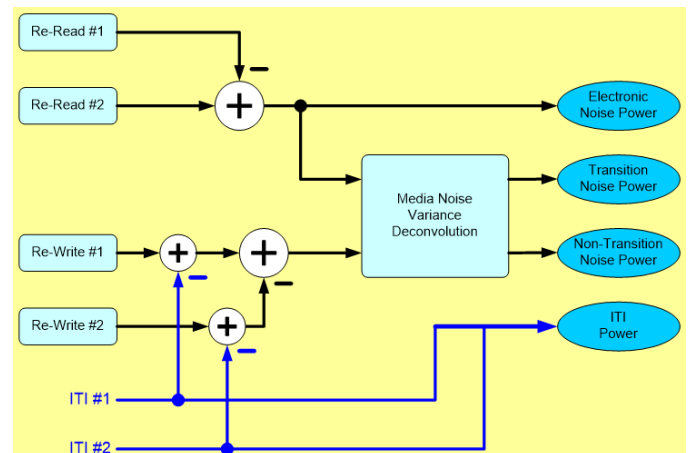


Fig. 2. Block diagram of noise component breakdown process

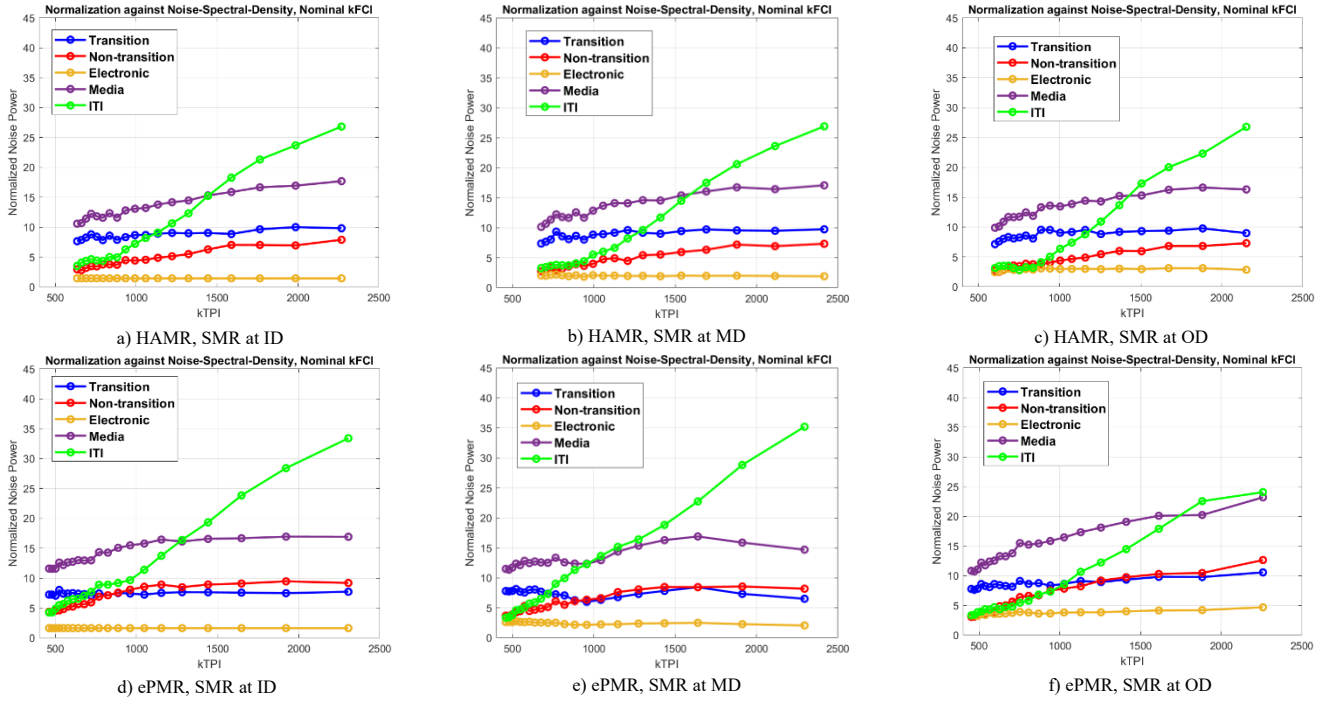


Fig. 3. Noise component breakdown for different environments

written. Re-Read #1 and Re-Read #2 are repeated reads of the same written data pattern; because the written medium is unchanged, their difference mainly estimates electronic noise power. The ITI #1 and ITI #2 paths estimate ITI from track N-1 and N+1. After subtracting the electronic noise power and ITI power from the total measured noise power, the remaining component is the media noise power, which is then separated into transition and non-transition components. Transition noise is associated with fluctuations near magnetic transitions, while non-transition noise represents fluctuations away from transition locations.

III. MEASUREMENT AND MODEL FITTING

Measurements were performed on contemporary HAMR and ePMR HDDs under SMR conditions. The media densities are 1.49, 1.57, and 1.27 Tbit/in² at ID, MD, and OD, and for the HAMR drive, are 1.23, 1.29, and 1.02 Tbit/in² at ID, MD, and OD, for the ePMR drive, respectively. The experimental conditions varied by adjusting the track density and linear density, represented by kilo-tracks per inch (kTPI) and kilo-flux changes per inch (kFCI). Readback waveforms were captured at 10 GigaSamples/s and processed for noise component analysis at ID, MD, and OD. The noise powers are normalized against the noise-spectral-density (NSD) at nominal kFCI for ID, where the NSD is referenced to the electronic or white noise floor and scaled by the corresponding data-rate or frequency-rate condition.

The SMR write order is also accounted for because it affects the adjacent track geometry seen by the reader. At OD, track N-1 is written before N, so N's leading edge later trims the edge of N-1 facing the center; therefore, N-1 contributes less ITI. Track N+1 is written after N, so its center-facing edge remains closer to the full natural write width and contributes more ITI. Thus, the effective fat track at OD is N+1. At ID and MD, the write

order reverses. In this case, N+1 is written before N and is trimmed, while N-1 is written after N and retains the fuller natural write width. Therefore, the effective fat track at ID and MD is N-1.

IV. RESULTS AND CONCLUSIONS

Fig. 3 presents the normalized noise-component results at nominal kFCI for six SMR cases: HAMR at ID, MD, and OD, followed by ePMR at ID, MD, and OD. Each panel reports transition, non-transition, electronic, media, and ITI noise power as a function of kTPI. Normalizing by NSD at nominal kFCI for ID places the HAMR/ePMR and ID/MD/OD measurements on a common signal processing basis. In Fig. 3, the electronic noise remains nearly flat after normalization, while ITI increases monotonically with kTPI. The media, transition, and non-transition noise also increase with kTPI but saturate at higher kTPI values. A key observation is that the overall range of normalized noise power is the same for both HAMR and ePMR across all three radii, indicating that both technologies exhibit comparable noise component levels under SMR operating conditions. For HAMR, OD exhibits the lowest ITI while MD shows the highest ITI; in contrast, for ePMR, the ITI levels are approximately the same across all three radii. The results show that decomposing total noise provides more useful insight than total-noise comparison alone. CMR data will be investigated for comparison, but those results are not shown in this paper.

REFERENCES

- [1] R. Wood et al., "Measured and Modeled Responses for Heat-Assisted Magnetic Recording Up to Ultra-High Areal Densities", *IEEE Trans. Magn.*, Vol. 62, No. 3, pp. 1-9, March 2026
- [2] J. Goode et al., "Pushing the Limits of Areal Density: Fusing Advanced Channel Coding, HAMR, and SMR in Next-Generation HDDs", *IEEE Trans. Magn.*, Vol. 62, No. 3, pp. 1-4, March 2026